

Health and Beauty Aids Department and Stationery Department

Market Basket Analysis Report

Introduction

In our analytical report for our client, a store that is interested in better understanding some of its transactions, we set out to identify associations between items purchased from the client's Health and Beauty Aids Department and Stationery Department using our client's transactional data. Upon our client's request, we will also be assessing whether or not sequence analysis can be performed on the provided dataset. Afterwards, we will perform market basket analysis on the transactional data and identify the two highest and two lowest lift values from our analysis. We will provide an accompanying interpretation of each of these rules. Finally, we will give practical recommendations to our client based on our analysis with the objective of helping to potentially increase our client's revenue.

Data Understanding

Figure 1 – Data Dictionary

Name	Model Rate	Measurement Level	Description
STORE	Rejected	Nominal	Identification number of the store
TRANSACTION	ID	Nominal	Transaction identification number
PRODUCT	Target	Nominal	Product purchased
QUANTITY	Rejected	Interval	Quantity of this product purchased

In this figure, we see a data dictionary for our dataset, which contains 459,258 data points. Each data point represents a transaction that was made over the past three months. We set our Product variable as our target variable, and our Transaction variable as our ID variable for the purpose of our analysis.

Figure 2 – List of Products

1. bar soap
2. bows
3. candy bars

4. deodorant
5. greeting cards
6. magazines
7. markers
8. pain relievers
9. pencils
10. pens
11. perfume
12. photo processing
13. prescription medications
14. shampoo
15. toothbrushes
16. toothpaste
17. wrapping paper

In the above figure, we list the different products, spanning across both our client's Health and Beauty Aids Department and Stationery Department, that are present in the dataset.

Can sequence analysis be performed on the given dataset? Why or why not?

Sequence analysis can not be performed on the given dataset. Sequence analysis typically is used to determine common sequences in time-ordered data and sequences associated with an event. However, our dataset is not time-ordered; we only know that certain items were bought together based on a mutual transaction ID. Therefore, because our dataset lacks data on the sequence of events, such as whether or not the purchase of a pen seemed to have sequentially led to the purchase of shampoo, we will have to use association analysis, and not sequence analysis.

Identifying the rules that have the two highest and two lowest lift values from our analysis, with an interpretation for each of these rules.

We began our analysis by importing our dataset into the SAS Viya software. Using SAS Studio, we generated code (Appendix Figure 1) to output three tables: one titled "ASSOC_RULES" (Appendix Figure 2) which lists association rules along with their support and count values, one titled "FREQMB" (Appendix Figure 3) which provides the support and count values for each of the 17 products, and one titled "MBA_RULES" which gives a comprehensive list of association rules along with accompanying results, such as count, support, confidence, and lift. The first two tables provided a degree of insight our analysis and dataset, but the third table most directly contributed to achieving our analytical objectives.

Figure 3 – MBA_RULES Table

	RULEID	LHS	RHS	COUNT	SUPPORT	CONF	LIFT	ITEM1	ITEM2	RULE
1	302	1	1	1215	0.6075	41.882109617	8.2137889032	Prescription Med	Wrapping Paper	Prescription Med ==> Wrapping Paper
2	366	1	1	1215	0.6075	11.914100804	8.2137889032	Wrapping Paper	Prescription Med	Wrapping Paper ==> Prescription Med
3	175	1	1	1655	0.8275	24.482248521	3.6350777314	Shampoo	Toothbrush	Shampoo ==> Toothbrush
4	239	1	1	1655	0.8275	12.286562732	3.6350777314	Toothbrush	Shampoo	Toothbrush ==> Shampoo
5	297	1	1	4364	2.182	24.255224544	3.6013696428	Perfume	Toothbrush	Perfume ==> Toothbrush
6	361	1	1	4364	2.182	32.397921307	3.6013696428	Toothbrush	Perfume	Toothbrush ==> Perfume
7	261	1	1	2268	1.134	20.752127368	3.0812364317	Bow	Toothbrush	Bow ==> Toothbrush
8	325	1	1	2268	1.134	16.837416481	3.0812364317	Toothbrush	Bow	Toothbrush ==> Bow
9	135	1	1	8732	4.366	25.53141721	1.7381909119	Candy Bar	Greeting Cards	Candy Bar ==> Greeting Cards
10	199	1	1	8732	4.366	29.72393369	1.7381909119	Greeting Cards	Candy Bar	Greeting Cards ==> Candy Bar
11	260	1	1	267	0.1335	2.4430414494	1.6842753874	Bow	Prescription Med	Bow ==> Prescription Med
12	324	1	1	267	0.1335	9.2037228542	1.6842753874	Prescription Med	Bow	Prescription Med ==> Bow
13	28	1	1	37	0.0185	2.29244114	1.5804489073	Markers	Prescription Med	Markers ==> Prescription Med
14	92	1	1	37	0.0185	1.2754222682	1.5804489073	Prescription Med	Markers	Prescription Med ==> Markers
15	146	1	1	5848	2.924	19.906729755	1.4753922368	Greeting Cards	Pencils	Greeting Cards ==> Pencils
16	210	1	1	5848	2.924	21.67129887	1.4753922368	Pencils	Greeting Cards	Pencils ==> Greeting Cards
17	139	1	1	7956	3.978	23.262477705	1.4500531529	Candy Bar	Toothpaste	Candy Bar ==> Toothpaste
18	203	1	1	7956	3.978	24.796633941	1.4500531529	Toothpaste	Candy Bar	Toothpaste ==> Candy Bar
19	264	1	1	6603	3.3015	19.306453028	1.4309025776	Candy Bar	Pencils	Candy Bar ==> Pencils
20	328	1	1	6603	3.3015	24.469149528	1.4309025776	Pencils	Candy Bar	Pencils ==> Candy Bar
21	5	1	1	518	0.259	4.7396834111	1.4022731986	Bow	Shampoo	Bow ==> Shampoo
22	69	1	1	518	0.259	7.6627218935	1.4022731986	Shampoo	Bow	Shampoo ==> Bow
23	134	1	1	759	0.3795	6.9448256931	1.3619975864	Bow	Wrapping Paper	Bow ==> Wrapping Paper
24	198	1	1	759	0.3795	7.4426358109	1.3619975864	Wrapping Paper	Bow	Wrapping Paper ==> Bow
25	21	1	1	6416	3.208	21.840215134	1.3613972345	Greeting Cards	Toothpaste	Greeting Cards ==> Toothpaste
26	85	1	1	6416	3.208	19.996883279	1.3613972345	Toothpaste	Greeting Cards	Toothpaste ==> Greeting Cards
27	49	1	1	775	0.3875	9.0063916328	1.3372519128	Soap	Toothbrush	Soap ==> Toothbrush
28	113	1	1	775	0.3875	5.7535263549	1.3372519128	Toothbrush	Soap	Toothbrush ==> Soap
29	41	1	1	783	0.3915	4.351934193	1.287554495	Perfume	Shampoo	Perfume ==> Shampoo
30	105	1	1	783	0.3915	11.582840237	1.287554495	Shampoo	Perfume	Shampoo ==> Perfume

In the above figure, we display the first 30 rows of our MBA_RULES table, which provides the count, support, confidence, and lift values for our association rules. The table is sorted by lift in descending order, in order to display the association rules with the highest lift values, and consists of 272 rows in total.

Figure 4 – Two Highest and Two Lowest Lift Values

	RULEID	COUNT	SUPPORT	CONF	LIFT	ITEM1	ITEM2	RULE
1	302	1215	0.6075	41.882109617	8.2137889032	Prescription Med	Wrapping Paper	Prescription Med ==> Wrapping Paper
2	366	1215	0.6075	11.914100804	8.2137889032	Wrapping Paper	Prescription Med	Wrapping Paper ==> Prescription Med
3	14	13	0.0065	1.1992619926	0.0835286082	Deodorant	Pens	Deodorant ==> Pens
4	78	13	0.0065	0.0452725057	0.0835286082	Pens	Deodorant	Pens ==> Deodorant

We took our MBA_RULES table and sorted by lift, in order to select just the two association rules with the highest lift values and the two association rules with the lowest lift values. In the above figure, we display the selected rules, with our first two rules having the highest lift values in our analysis and our last two rules having the lowest lift values in our analysis, along with the accompanying rule ID, count, support, confidence, and lift. Our first

rule, Prescription Med $\rightarrow$ Wrapping Paper, has a lift value of roughly 8.2, indicating that the rule is much better for prediction than a random guess, and that there is a strong association between the two items; this indicates a relatively higher probability of these items being purchased together in comparison to the items having no association with each other. Similarly, our second rule, Wrapping Paper $\rightarrow$ Prescription Med, also has a lift value of roughly 8.2, indicating that the rule is much better for prediction than a random guess and signifying a strong association between the two items. This likewise indicates a relatively high probability of these items being purchased together as opposed to the items being independent of each other in transactions.

Our third rule, Deodorant $\rightarrow$ Pens, has a lift value of roughly 0.08, far below the standard value of 1 in order for an association rule to be considered useful insight into the association between two items or item sets. This low lift value indicates that the rule is not much better for prediction than a random guess, and that there is a weak association between the two items; thus, there is not much association between the purchase of deodorant and the purchase of pens. Likewise, our fourth rule, Pens $\rightarrow$ Deodorant, also has a low lift value of roughly 0.08, indicating that the rule is not much better for prediction than a random guess, and that there is a weak association between the purchase of the two items.

We note that the two rules with the highest lift values, Prescription Med $\rightarrow$ Wrapping Paper and Wrapping Paper $\rightarrow$ Prescription Med, have the same lift value of roughly 8.2, and consist of the same items, albeit in different order. Correspondingly, we note that the two rules with the lowest lift values, Deodorant $\rightarrow$ Pens and Pens $\rightarrow$ Deodorant, have the same lift value of roughly 0.08, and also consist of the same items. These observations are in line with the nature of our dataset: unlike datasets that are viable for sequence analysis, our dataset does not consist of time-ordered data or sequences. For example, both the rules Prescription Med $\rightarrow$ Wrapping Paper and Wrapping Paper $\rightarrow$ Prescription Med are identical; they both simply refer to the two items being purchased together, and not in any sequential order. Therefore, it makes sense that the rules with the two highest lift values are simply differently ordered versions of the same association, and that the rules with the two lowest lift values are differently ordered versions of the same association.

Recommendations

Market basket analysis is a powerful tool in order for businesses to understand patterns in customer buying behavior and to make subsequent business decisions. The lift values that accompany the association rules indicate the strength of association between items or item sets, and therefore give insight into which items are more likely to be purchased together. In general, a lift value of at least one indicates a decently strong association between the items involved in the association rule.

We would therefore give the following recommendations for the items that are associated together in association rules with lift values of at least 1. The store should either place these items closer together in the store, to make it more convenient for customers to purchase items that seem to be more popularly purchased together, or place the items further apart in the store, to encourage customers to see more products while walking between the items and thus to potentially purchase more products. The associated items could be packaged together to encourage more customers to purchase these items together, or the associated items could be packaged with poorly selling items, in order to increase the sales of poorly selling items. Not advertising the strongly associated items together could also be strategically advantageous from a marketing perspective; marketing resources could be saved by simply marketing one of the items, since there is a relatively high probability that the non-advertised item will have higher sales if the advertisements increase the sales of the advertised item.

Lastly, we would recommend that additional data be collected for higher-quality market basket analysis in the future. Data collected over years, rather than over just three months, should provide greater insight into the association between items, and allow for more robust association rules to be generated. The inclusion of customer demographic data could help to target particular customer groups with advertisements and thus increase the efficiency of marketing efforts, and the usage of training and validation data could be used to test the validity of the association rules.

We wish our client all the best in their marketing efforts, and we hope that our analysis and recommendations will be helpful to the sales and marketing teams in contributing to the financial success of our client.

Appendix

Figure 1 – SAS Studio Market Basket Analysis Code

```
1  cas mysession;
2
3  libname mycaslib cas caslib='casuser';
4
5  Ⓣ proc mbanalysis data=mycaslib.TRANSACTIONS pctsupport=0 conf=0 lift=0;
6      id transaction;
7      target product;
8      output out=mycaslib.assoc_rules
9             outfreq=mycaslib.freqmb
10            outrule=mycaslib.mba_rules;
11  run;
```

Figure 2 – ASSOC_RULES Table

	size	count	support	Item1	Item2
1	2	8732	4.366	Candy Bar	Greeting Cards
2	2	8107	4.0535	Candy Bar	Magazine
3	2	7956	3.978	Candy Bar	Toothpaste
4	2	7267	3.6335	Greeting Cards	Magazine
5	2	6603	3.3015	Candy Bar	Pencils
6	2	6416	3.208	Greeting Cards	Toothpaste
7	2	6333	3.1665	Magazine	Toothpaste
8	2	6326	3.163	Magazine	Pencils
9	2	5848	2.924	Greeting Cards	Pencils
10	2	4912	2.456	Pencils	Toothpaste
11	2	4364	2.182	Perfume	Toothbrush
12	2	4230	2.115	Magazine	Pens
13	2	3438	1.719	Magazine	Perfume
14	2	3291	1.6455	Magazine	Photo Processing
15	2	2635	1.3175	Magazine	Toothbrush
16	2	2453	1.2265	Candy Bar	Pens
17	2	2268	1.134	Bow	Toothbrush
18	2	2063	1.0315	Pens	Toothpaste
19	2	1670	0.835	Greeting Cards	Pens
20	2	1655	0.8275	Shampoo	Toothbrush

The table was sorted by support in descending order. Only the first 20 of 136 rows are shown here.

Figure 3 – FREQMB Table

	item	count	support
1	Magazine	48261	24.1305
2	Candy Bar	34201	17.1005
3	Toothpaste	32085	16.0425
4	Greeting Cards	29377	14.6885
5	Pens	28715	14.3575
6	Pencils	26985	13.4925
7	Perfume	17992	8.996
8	Toothbrush	13470	6.735
9	Photo Processing	11696	5.848
10	Bow	10929	5.4645
11	Wrapping Paper	10198	5.099
12	Soap	8605	4.3025
13	Shampoo	6760	3.38
14	Pain Reliever	5340	2.67
15	Prescription Med	2901	1.4505
16	Markers	1614	0.807
17	Deodorant	1084	0.542

The table was sorted by support in descending order.